

Literature Review Evidence Table

Hardware-Aware Transformer Inference on CUDA and Apple Metal

Scope

This PDF synthesizes all 17 individually identifiable CUDA/Metal works or sources cited in the thesis DOCX and PPTX materials. The review is organized by performance mechanism: workload foundations, attention and memory efficiency, platforms and profiling, serving, and evaluation.

How to read the table

Each row states the source's main contribution, the evidence it contributes to the thesis, and the role it should play in the proposed controlled portability study. Citation statuses identify sources that need verification before final submission.

Scope limitation

The supplied Sample Literature.pdf concerns IoT/MEC offloading and contains numbered citations without a bibliography in the provided extract. It is listed in the source inventory but is not mixed into this CUDA/Metal literature synthesis.

Literature review evidence table — sources R01-R02 (1/9)

Reference	Main contribution / evidence	Role in thesis	Source and citation status
R01 Workload foundations Vaswani et al. (2017). Attention Is All You Need. NeurIPS.	Defines the Transformer and scaled dot-product self-attention.	Provides the attention workload foundation; report prefill and decode separately.	Peer-reviewed. Found in DOCX and PPTX.
R02 Workload foundations Devlin et al. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. NAACL.	Shows the Transformer as a reusable large-scale language-model workload.	Justifies realistic Transformer inference evaluation and model selection.	Peer-reviewed. PPTX only; citation completed from standard metadata.

Yellow rows need citation completion or verification before thesis submission.

Literature review evidence table — sources R03-R04 (2/9)

Reference	Main contribution / evidence	Role in thesis	Source and citation status
R03 Workload foundations Sze et al. (2017). Efficient Processing of Deep Neural Networks: A Tutorial and Survey. Proceedings of the IEEE, 105(12), 2295–2329.	Surveys data reuse, energy, throughput, hardware, and evaluation trade-offs.	Supports measuring memory, latency, and throughput together rather than FLOPs alone.	Peer-reviewed. DOCX and PPTX.
R04 Platforms and profiling Shen et al. (2018). CUDAAAdvisor: LLVM-Based Runtime Profiling for Modern GPUs. CGO, 214–227.	Profiles runtime and memory behavior across CUDA versions and GPU architectures.	Use profiler evidence to attribute a speedup; do not rely on elapsed time alone.	Peer-reviewed. DOCX and PPTX.

Yellow rows need citation completion or verification before thesis submission.

Literature review evidence table — sources R05-R06 (3/9)

Reference	Main contribution / evidence	Role in thesis	Source and citation status
R05 Platforms and profiling Mittal & Vaishay (2019). A Survey of Techniques for Optimizing Deep Learning on GPUs. Journal of Systems Architecture, 99, 101635.	Classifies architecture- and system-level GPU optimization methods.	Provides a CUDA taxonomy for interpreting memory, precision, computation, and parallelism.	Peer-reviewed. DOCX and PPTX.
R06 Attention and memory efficiency Dao et al. (2022). FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness. NeurIPS.	Uses tiling and fusion to reduce HBM↔SRAM traffic while preserving exact attention.	Primary portable principle: reduce high-bandwidth-memory reads and writes.	Peer-reviewed. DOCX and PPTX.

Yellow rows need citation completion or verification before thesis submission.

Literature review evidence table — sources R07-R08 (4/9)

Reference	Main contribution / evidence	Role in thesis	Source and citation status
R07 Attention and memory efficiency Dao (2024). FlashAttention-2: Faster Attention with Better Parallelism and Work Partitioning. ICLR.	Improves parallel work partitioning beyond the first FlashAttention kernel.	Compare the IO-aware principle with backend-specific execution mapping.	Peer-reviewed. DOCX and PPTX.
R08 Attention and memory efficiency Shah et al. (2024). FlashAttention-3: Fast and Accurate Attention with Asynchrony and Low-precision. NeurIPS.	Uses Hopper-specific asynchronous pipelines, warp specialization, and low precision.	Key portability-limit example: later gains rely on NVIDIA-specific facilities.	Peer-reviewed. DOCX and PPTX.

Yellow rows need citation completion or verification before thesis submission.

Literature review evidence table — sources R09-R10 (5/9)

Reference	Main contribution / evidence	Role in thesis	Source and citation status
R09 Attention and memory efficiency Yuan et al. (2025). Native Sparse Attention: Hardware-Aligned and Natively Trainable Sparse Attention. ACL.	Combines hierarchical token compression, selection, and local windows with hardware-aligned sparsity.	For long context, measure quality and phase-specific latency, not only FLOPs.	Peer-reviewed. DOCX and PPTX.
R10 Attention and memory efficiency Alizadeh et al. (2024). LLM in a Flash: Efficient Large Language Model Inference with Limited Memory. ACL.	Selectively loads model parameters from flash storage into memory on demand.	Keep parameter-residency methods distinct from attention-kernel methods.	Peer-reviewed. DOCX and PPTX.

Yellow rows need citation completion or verification before thesis submission.

Literature review evidence table — sources R11-R12 (6/9)

Reference	Main contribution / evidence	Role in thesis	Source and citation status
R11 Serving Kwon et al. (2023). Efficient Memory Management for Large Language Model Serving with PagedAttention. SOSP.	Pages the KV cache to reduce fragmentation and improve sharing and batching.	Measure isolated kernels and end-to-end serving separately; report cache and batching conditions.	Peer-reviewed. DOCX and PPTX.
R12 Platforms and profiling Benazir & Lin (2025). Profiling Large Language Model Inference on Apple Silicon: A Quantization Perspective. arXiv:2508.08531.	Profiles Apple Silicon and NVIDIA across model sizes and quantization settings.	Use system-level Metal profiling and state the precision and execution path explicitly.	Preprint. DOCX and PPTX; local DOCX venue claim requires verification.

Yellow rows need citation completion or verification before thesis submission.

Literature review evidence table — sources R13-R14 (7/9)

Reference	Main contribution / evidence	Role in thesis	Source and citation status
R13 Platforms and profiling Kirk & Hwu (2022). Programming Massively Parallel Processors: A Hands-on Approach (4th ed.). Morgan Kaufmann.	Explains coalescing, tiling, synchronization, occupancy, and CUDA execution.	Use as background for CUDA diagnosis; not direct evidence of Metal portability.	Textbook. PPTX only; citation completed from standard metadata.
R14 Platforms and profiling Apple ML Research (2025). MLX / Apple Silicon platform evidence (exact report unspecified in local files).	Materials describe unified memory, Metal Performance Primitives, and distinct prompt/generation regimes.	Use only as contextual platform evidence; keep it distinct from independent studies.	Incomplete. DOCX and PPTX; add exact report, author, date, URL, and access date.

Yellow rows need citation completion or verification before thesis submission.

Literature review evidence table — sources R15-R16 (8/9)

Reference	Main contribution / evidence	Role in thesis	Source and citation status
R15 Serving and deployment Wu et al. (2019). Edge-inference deployment study (title and venue unspecified in local files).	Cited for operator support and hardware fragmentation in edge deployment.	Document runtime/backend coverage; distinguish Metal GPU, Core ML, and ANE paths.	Incomplete. DOCX and PPTX; locate original title, venue, and identifier.
R16 Serving and evaluation MLCommons / MLPerf Inference benchmark suite (exact version unspecified in local files).	Defines scenario-, quality-, version-, and run-rule-based performance evaluation.	Declare model, precision, shapes, versions, warm-up, repetitions, scenario, and quality constraints.	Incomplete. DOCX and PPTX; cite the precise MLPerf Inference version and ruleset.

Yellow rows need citation completion or verification before thesis submission.

Literature review evidence table — sources R17-R17 (9/9)

Reference	Main contribution / evidence	Role in thesis	Source and citation status
R17 Serving and evaluation Schryen (2024). Speedup and Efficiency of Computational Parallelization: A Unifying Approach and Asymptotic Analysis. JPDC, 185, 104801.	Unifies speedup and efficiency models and analyzes overhead and scalability.	Report within-platform relative speedups and explain residual serial and system overhead.	Peer-reviewed. DOCX and PPTX.

Yellow rows need citation completion or verification before thesis submission.

Source inventory and coverage

Local thesis-directory file	Coverage decision
Tursyn_CSCI693_Literature_Review_Introduction_	Primary prose draft; selected-reference set.
Tursyn_CSCI693_Literature_Review_Introduction.	Near-duplicate prose draft; no additional identifiable sources.
Tursyn_CSCI693_Literature_Review_Introduction2	Earlier draft; adds incomplete references to Devlin, Apple MLX, Wu, and MLPerf.
Tursyn_CSCI693_Thesis_Literature_Review_Intro	Presentation; adds Kirk & Hwu and confirms the mechanism-based review structure.
Sample Literature.pdf	Unrelated IoT/MEC-offloading excerpt. It has numbered citations but no supplied bibliography; excluded from this CUDA/Metal synthesis.

Coverage total: 17 individually identifiable CUDA/Metal works or sources (R01-R17).

Citation checks required before thesis submission

Reference	Issue	Required action
R12	Local files name a PACM MACS venue; the identified 2025 work is an arXiv preprint.	Confirm a final peer-reviewed version, if any, before submission.
R14	Apple MLX/M5 platform evidence lacks title, author, publication date, and URL.	Replace with the exact Apple report and an access date.
R15	Wu et al. lacks title, venue, and persistent identifier.	Locate and enter a complete APA 7 citation before relying on its claim.
R16	MLPerf Inference lacks a version, benchmark paper, and ruleset.	Cite the exact version and scenario used to guide the experiment.
Sample PDF	The IoT/MEC sample has no reference pages in the supplied extract.	Obtain its bibliography before making a separate IoT-offloading table.